

HIEU TRAN

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Hanoi, Vietnam

Education

Hanoi Amsterdam High School for the Gifted, Hanoi, Vietnam

2023 - 2026

Specialization: **Informatics**

GPA Grade 11: **9.5**

GPA Grade 10: **9.3**

Standardized Tests

SAT: **1570**

AP Computer Science: **5**

AP Calculus AB: **5**

IELTS: **7.5**

Technical Skills

Competitive Programming: C++20, Python | Codeforces Master (*top 3% globally*)

AI/Machine Learning: PyTorch, Lightning, FastAPI, Docker, ONNX, Redis

Full-Stack Development: React 18, TypeScript, Next.js, TailwindCSS, PostgreSQL

Tools & Workflow: Git, Docker Compose, PyTest, Jest, GitHub Actions

Achievements & Honors

Title of Outstanding Student of the Capital, Hanoi Capital of Vietnam

2025

Champion, Vietnam AI Championship

2025

Champion, Vietnam National Young Informatics Contest

2025

Second Prize (*Ranked top 10/200*), **Vietnam Olympiad in Artificial Intelligence**

2025

Third Prize, Vietnam National Informatics Olympiad

2025

Silver Medal, World Invention Creativity Olympiad (WICO)

2024

Third Prize (*Asia Round*), **International Collegiate Programming Contest**

2024

First Prize in High School Division (*National Round*), **International Collegiate Programming Contest**

2024

First Prize in Scientific Product, **Hanoi Young Informatics Contest**

2024

Second Prize, Hanoi Informatics Olympiad for grade 12

2024

Gold Medal, Vietnam High School for Gifted Students Olympiad

2024

Gold Medal & Talent Award, **Central Highlands Informatics Olympiad**

2024

Silver Medal, Vietnam National University Olympiad

2024

Top 3 Vietnam, Microsoft Imagine Cup Junior

2024

Research Initiatives & Invention

Co-Inventor, Lead Programmer & Developer, Littlehands | [Github](#) | [Video Introduction](#)

The application platform focuses on supporting people with hearing impairments to communicate effectively

- Collected data from 30+ teachers and students with hearing impairments. Recorded sign language with 100 vocabulary entries
- Labeled 84,000+ multi-dimensional vocabulary videos (3s-4s/video) on the Labellerr system and manually screened on 200 outliers or poor quality videos
- Trained computer vision models with Video Transformer Network to recognize sign language gestures. Fine-tuned NLP models (BERT, GPT) with prompt engineering to convert Vietnamese text into sign language syntax
- Developed PostgreSQL backends and implemented FastAPI (Python) to connect the user interface with deep learning models
- Used C# language to optimize an intuitive, minimalist, and deaf-friendly interface

Programmer & Developer, Sign Helper | [Coding](#)

A web platform focused on helping deaf people receive information effectively, Littlehands' prototype

- Built-in OCR and Speech-to-Text APIs to convert text, audio, and video format information into sign language

Co-author, Sign Language Recognition: Multi-view Dataset & Evaluation | [PDF](#) | [Github](#)

Instructor: Assoc. Prof. Phi Le Nguyen, Executive Director, Center for Artificial Intelligence, Hanoi University of Science and Technology

- Created a large-scale multi-view sign language dataset with 1,000 glosses, 30 signers, and 84,000+ videos
- Evaluated state-of-the-art sign language recognition models on a large-scale dataset through extensive experiments
- Indicated that utilizing multi-view data substantially enhances model accuracy across all models with a maximum performance improvement of up to 19.75% compared to models trained on single-view data
- Published on the website of Winter Conference on Applications of Computer Vision (WACV) [\[Link\]](#)

Author & Researcher, Optimized Paraphrase Detection with Multi-Architecture Ensembles | [PDF](#)

- Developed and optimized a paraphrase detection model using Siamese networks, Transformer encoders, and FastText embeddings, increasing accuracy from 0.75 to 0.82
- Created a multi-architecture ensemble combining Transformer and ESIM models, achieving a final accuracy of 0.89 through K-Fold Cross-Validation and advanced training techniques

Author & Researcher, A Comprehensive Survey of Square Root Decomposition | [PDF](#)

- Conducted a survey on Square Root Decomposition, including the development of advanced optimizations like Mo's Algorithm, Hilbert Curve Sorting, and Heavy-Light decomposition for efficient query handling
- Demonstrated practical applications and optimizations through case studies and real-world benchmarks. Led the design and implementation of performance tuning strategies

Peer Reviewer, SPARK | [Website](#)

A student platform for academic research & knowledge

- Reviewed 5+ high school research papers on IoT, AI, and LED applications in smart agriculture, demonstrating expertise in sustainability, data analysis, and ML-driven optimization of crop productivity
- Delivered constructive feedback to help improve structure, argumentation, and scholarly rigor within strict deadlines [\[Link\]](#)

Water Puppet Spreadings

Founder & Operator, Moc Roi - Moc Water Puppet | [Store](#)

A water puppet products startup to promote Vietnamese culture with 50% of the profits donated to deaf children communities in Vietnam

- Contacted and cooperated with 05 artisans in Ward Village Water Puppet Dao Thuc (Hanoi), 55 Hang Gai (Hoan Kiem, Hanoi) and Nghia Hung (Nam Dinh) to build a stable supply of products. The first batch of imports was 6,000 products at a cost of about \$11,000
- Partnered with 10 deaf children to produce water puppets with an average output of 250 products/quarter
- Deployed a Shopee store and opened 10 short-term business booths at tourist streets and cultural events in Hanoi with a revenue from retail sales is \$7,500
- Cooperated and authorized to sign 03 product supply contracts with a total revenue of \$24,500 with 3 large travel companies in Vietnam to serve business for foreign tourists

Craft Teacher & Advertising Freelancer, Ward Village Water Puppet Dao Thuc | [Report](#)

The oldest traditional village in Vietnam specializing in crafting and performing water puppetry (300 years), recognized as a National Intangible Cultural Heritage and performed across Vietnam and internationally (USA, France, the United Kingdom, Germany, Japan,...)

- Leading foreign and domestic tourists with 1 week/trip to promote the value of 300 years of Vietnamese water puppetry culture through the experience of the stage hall and puppet workshop at Ward Village Water Puppet Dao Thuc
- Opening 20 monthly permanent workshops to guide visitors to color and polish raw water puppets, and guiding the experience of entering the water to control the puppets directly
- Teaching 10 deaf students 1 session/week for 03 months the basic techniques and skills of making and sculpting wooden chicken and wooden duck, and how to use wood chisels and scrubbers,... at Hanoi School for Deaf and Dumb Children
- 2-hour presented on the structure of wooden water puppets and donated 10 water puppet characters to 20 children of Agent Orange in Da Nang

Director & Performer, Art Show “Thanh Am Ban Nguyet - Harmonies of the Crescent Lake” | [Fanpage](#)

An 1-hour free water puppet show held annually including 15 water puppet performances adapted from Vietnamese folk tales (farming, animal husbandry, and festivals,...) at Ward Village Water Puppet Dao Thuc to spread the value of Vietnamese art form mainly to the deaf

- Managed 20 members with the roles of Sound, Lighting, Performance, Stage, Audience Management, Sign Interpretation
- Administered \$15,500 fund including 80% from the Moc Roi project & Vietnam Puppet Theatre and 20% from individuals & the Hanoi high schools' Parent Associations
- Contacted, recruited and selected 08 water puppeteers, musicians and MCs with 15+ years of experience working in the profession. Arranged, assigned roles and approved scripts for each play
- Organized performance rehearsals for 01 month and reviewed 02 times, detecting and improving 95% of errors in matching sound - light - timestamp performances. Coordinated with sign language interpreters in rehearsals and script matching
- Directly performed in 03 water puppetry performances with a total duration of 20 minutes
- Invited 50 children with disabilities and 30 foreign tourists to attend, receiving 100% positive feedback on cultural values and art genres that are both suitable and psychologically healing for the deaf

Technological Education

Vice President & Head of Expertise, Illuminating Project | [Fanpage](#)

A STEM and Code Teaching Project for deaf children in Vietnam encouraging the message "No One Is Left Behind When The Light Is Lit"

- Coordinated 40 members and volunteers and managed 10 teams of mentors and students when developing robotics project ideas. Organized and trained weekly lessons on robotics and VR programming for 50+ children with hearing impairments
- Led the Month of STEM events at the Hanoi School for Deaf and Dumb Children, including side events such as Robotics Assembly Workshop for Grade 1 children, Robotics Workshop with speakers from Hanoi University of Science and Technology, etc. Hosted the main event “Robotics Tournament” of STEM Month with the competition of 10 teams of mentors and deaf students at an organizing cost of \$750

President, Accessible Technological Engineering Club | [Fanpage](#)

A research and transmission project of assistive technologies for the disabled community

- Researched and analyzed a project focused on mobility support for people with brain injuries using sensor technology
- Hosted the "Exoskeleton" webinar with speakers from MIT and the University of Chicago, discussing the disability assistive technology market attended by 100+ high school students nationwide
- Maintained and developed monthly AI programming lesson plans for 10+ poor high school students in the northern highland provinces of Vietnam
- Led LEGO robotics workshops, guiding assembly and block-based programming, and facilitated a robot racing minigame for 50+ children at Da Nang Center for Supporting Agent Orange Victims and Disadvantaged Children

Tutor & Collaborator, HNCode | [Fanpage](#)

A prestigious organization in Hanoi specializing in providing programming courses from basic to advanced

- Teaching 5 students in grade 9 and 5 students in grade 10 programming in C++. Helped 4 students achieve high results in the city-level informatics olympiad

Internship

Algorithm Optimizer Intern, AimeOpt of Aimesoft | [Website](#) | [Certificate](#)

The first company to develop multimodal AI with 200+ installations worldwide

- Processed data from e-tickets, gas sensors, cameras, IoT, and weather data
- Built a standard data model for the train network (route, trip, station, stop time, ke). Deployed real-time data processing infrastructure and store history using pipeline streaming
- Adjusted train schedules and speeds in real-time. Optimized running time, train spacing, car allocation, and stopovers using heuristic algorithms. Developed personalized passenger route suggestions and simulated scenarios with digital twin and Monte Carlo methods for performance testing.

Talents

Vietnamese Arts: Water Puppet Performing, Water Puppet Crafting

Sports: Swimming, Soccer, Badminton, Table Tennis